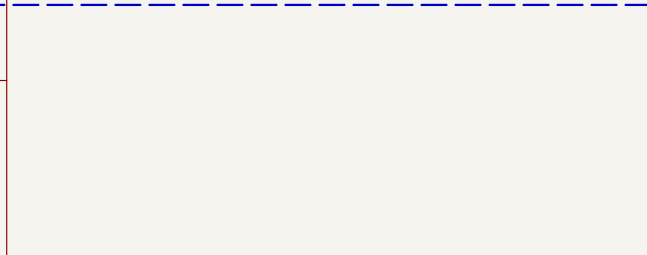


CC1/CC2 direct to CH224K (internal Rd). DP/DM of CH224K shorted together and NOT connected to the connector (WCH 'PD-only' mode) so USB D+/D- go to the MCU. CFG2/CFG3 float = single-resistor mode. CFG1 56k = 15V. PG low = contract OK. CH224K VBUS sense pin left NC (allowed in PD-only mode; pin is only 13.5V rated).



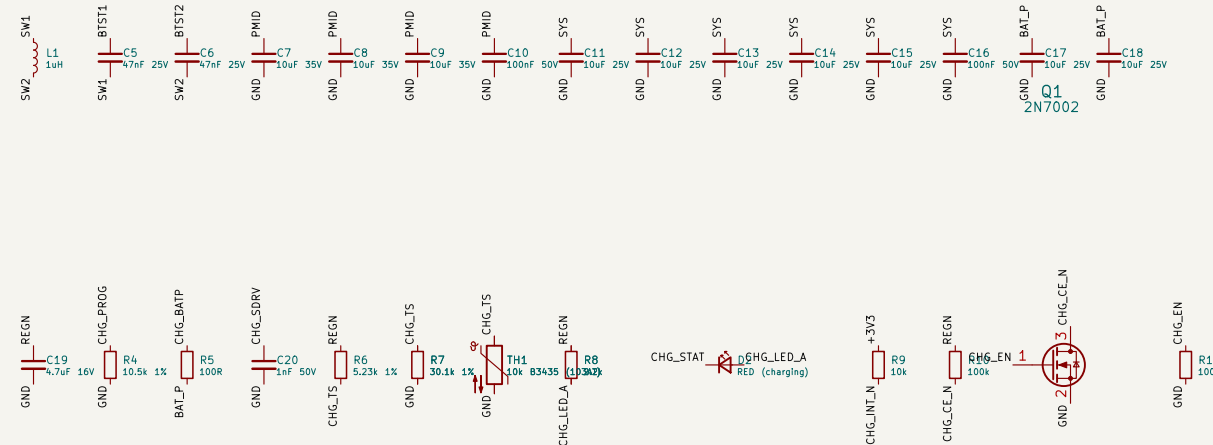
3V3 = OR of: VBUS (charger OR a phone in the USB-C port) | battery via Q2 when the MCU asserts PWR\_HOLD (lets the MCU finish balancing, put BQ25798 in shutdown + BQ76920 in SHIP after unplug, then release itself). Nothing else can power the MCU from the pack. Idle battery drain ~1-2uA total.



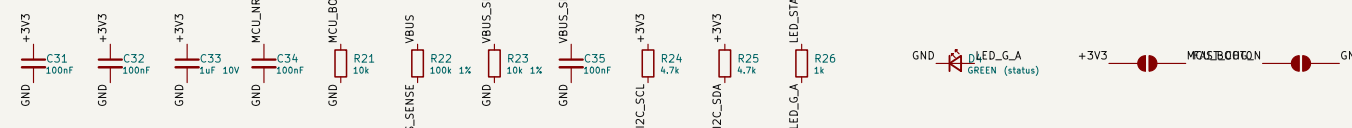
```

PROG 10.5k -> 3S / 1.5MHz; POR defaults VREG=12.6V, ICHG=1A, VYSMIN=9V.
NO ACFTETS: VAC1/VAC2 tied to VBUS, ADCRV1/2 to GND, ILIM_HIZ->RGEN (limit set by i2C)
/CE held HIGH by 100k -> NO charging until MCU drives CHG_EN high (after cell check).
BQ25798 D+/D- left open: MCU must clear AUTO_INDET_EN and set INPDM open i2C.
TS: 5.23k/30.1k + 10k NTC = 0.60C window (datasheet 7.3.9.5). Shutdown mode = 0.5uA.

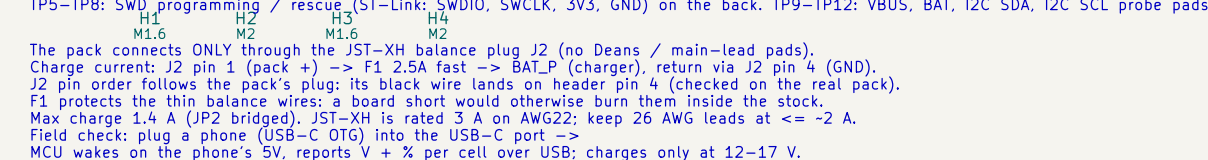
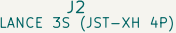
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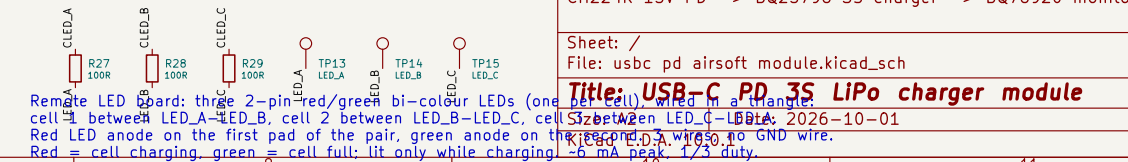
PF0/PF1 = I2C1 SDA/SCL (AF1) PA11/PA12 remapped onto pins 17/18 for USB (SYSCFG PA11-PA12\_RMP).  
PA1/PA3/PA0 = charlieplexed cell LEDs A/B/C (pads TP13-15). Charger /INT, CH224K PG and BMS ALRP are not  
wired to the MCU (firmware pins I2C). PA2 CHG\_EN, PA4 BMS boot, PA5 VBUS/11 ADC, PA6 LED, PA7 PWR\_HOLD.  
PB1 = FAST\_CHG\_EN (JP2 to GND, internal pull-up): open = slow charge (default), bridged = fast charge.  
Bridge JP1 + plug USB = ST ROM DFU bootloader. Backup: ST-Link on test pads TP5 SWDIO, TP6 SWCLK, TP7 3V3, TP8 GND.



3S per datasheet Table 9-2: VC5-VC4 = cell13, VC4=VC3=VC2 shorted, VC2-VC1 = cell2, VC1-VC0 = cell1.  
Rc=56R / Cc=1uF cell filters --> ~37mA internal balancing (datasheet min Rc 40R, max 50mA) (never balance adjacent cells together)  
SRP/SRN to VSS (no coulomb counter), CHG/DSG unused. Sits in SHIP mode (~0.6uA) when idle.  
Balance lead also carries the charge current (pins 1 and 4): firmware pauses charging before each cell reading.



| FEED_A | FEED_B | FEED_C | CH22 IR 1641.65 $\text{cm}^{-1}$ BQ25750 SS energizer $\times$ BQ76520 Inhibitor |
|--------|--------|--------|----------------------------------------------------------------------------------|
|        |        |        |                                                                                  |



Sheet: 1  
File: usbc\_pd\_airsort\_module.kicad.sch  
Title: USB-C PD 3S LiPo charger module  
Remate LED board: three 2-pin red/green bi-colour LEDs (one per colour) wired in a triangle.  
Red LED anode to LED1 pin 2, cathode to LED2 pin 2, between LED1 and LED2.  
Red LED anode on the first pad of the pair, green anode on the second pad of the pair, both wires to GND wire.  
Red = cell charging, green = cell full; lit only while charging. - 6 mA peak, 1/3 duty.